Handoff 03: Course catalog interface (stage 3)

Date: 2026-10-25 (America/Toronto)
Written by: Claude Code (implementer)
For: the human maintainer, and ChatGPT or Claude chat (reviewer) without repository access
Status: **in_review**: pull request open, not merged, not deployed

> **How to use this file:** upload it with `build.md` (and handoffs 01–02 if the chat is new). It summarizes evidence; the pull request diff and CI results are the source of truth (`build.md` section 14).

Starter prompt for the chat assistant

```
You are reviewing Python Academy stage 3: the Khan Academy-style course
catalog interface. I'm attaching build.md and this handoff, including
screenshots. Check it against the acceptance criteria below. Focus on:
is anything misleading to a learner (for example, could a coming-soon
course look startable); does progress shown match what was earned;
accessibility and mobile use; and what is not yet verified. Explain at a
first-year computer engineering level, and only call something tested if
this handoff shows the evidence.
```

1. Where we are

Stage	What	Status
Plan	`docs/plan/course-catalog.md`	Merged (PR #6)
1. ENG-01	Content format and result states	Merged (PR #5), verified live
2. ENG-02	Migrations, XP and badge ledgers, backfill	Merged (PR #7); maintainer confirmed live at migration 0003
3. Catalog UI	Backfill re-run, read endpoints, catalog/course/lesson pages	**in_review**: this PR
4. Goals and recommendations	Goal card and rule-based recommendation function	Not started

Rules for every stage: one branch and PR; exact test commands and results; never merge or deploy; no secrets or paid services; no change to age rules or account policy; never run learner code on the server; stop for review.

2. Handoff record (build.md template)

Field	Value
Task	Stage 3: catalog interface (plan sections 3.3, 7.5 stage 3; decision 8 "Coming soon")
Human goal	Let learners see every course, what is available, and their progress, and move through lessons with a clear sidebar, without changing how a lesson teaches
Repository and branch	`torontojohn/python-academy`, branch `ui/catalog`
Base commit / code commits	Base `60564e5` (`main` after PR #7) / `9eb3b2d` (backend), `5150a63` (interface)
Deployment status	Not requested. Not merged, not deployed

What changed

Backend (done first, as asked)

- **Migration** `0004` re-runs the idempotent award backfill. This closes the gap noted in handoff 02: a pass saved by the old app in the seconds of PR #7's deploy now gets its award before anything reads the ledger.
- `GET /api/catalog` (public): every course, unit, and lesson summary, with release status and prerequisites. Frozen in `backend/tests/fixtures/catalog_v1.json`, which the browser tests also use.
- `GET /api/awards` (signed-in students only): total XP, completed lessons, and badges, in catalog order.
- `/api/progress` **and the parent's "quests completed"** now read from the XP ledger. The response shape is unchanged. The parent count is now consistent with badges: it previously counted legacy arithmetic attempts as completed quests.

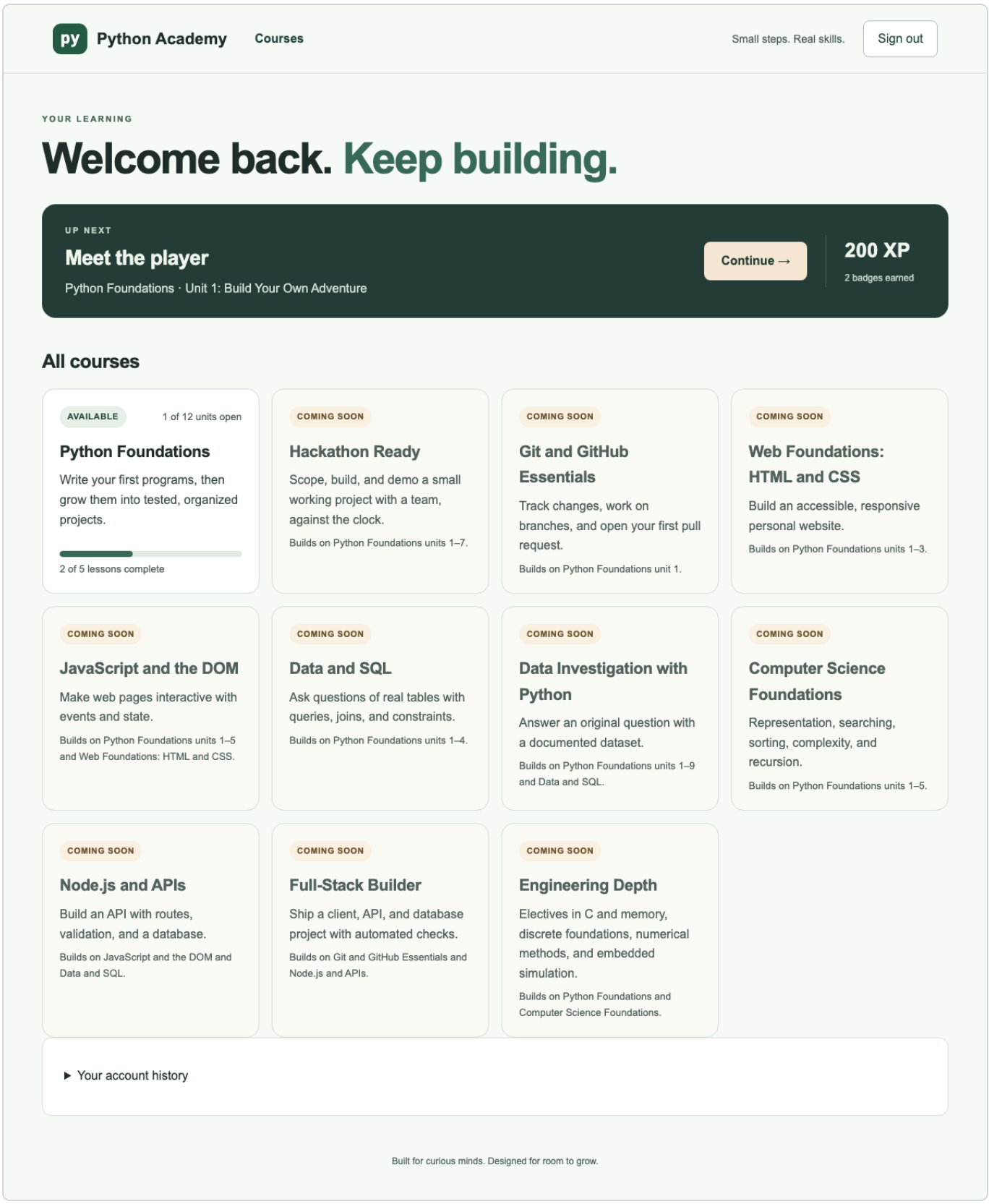
Interface

- `/courses`**, catalog:** one card per course with its title, summary, status chip ("Available" or "Coming soon"), and, for students, a progress bar ("2 of 5 lessons complete"). Coming-soon cards say what they build on, for example "Builds on Python Foundations units 1–7".
- `/courses/<id>`**, course page:** breadcrumb; title and summary; "Builds on" and "Helpful first"; progress and a Continue button; then each unit with its lessons marked Completed, Up next, or Not started. Coming-soon units show a label and no links. A coming-soon course says "This course is coming soon and cannot be started yet" and has **no start button**.
- `/learn/<lesson>`**, lesson page:** the same one-card-at-a-time flow, with a **unit sidebar** listing the unit's lessons as done (✓), current (highlighted, `aria-current="page"`), or not started, plus unit progress. Below 1100px wide it becomes a **collapsible menu** ("Lesson 3 of 5 · 2 complete"). Breadcrumbs link back to the course and catalog.

- **Student home (`/`):** "Welcome back. Keep building." with an Up next card (Continue button, XP, badges), then all courses.
- **Parents and visitors** open lessons as a preview that saves nothing, as before. Parents get a "Browse all courses" link.
- **Shared header** with a Courses link; fixed to fit a 375px phone.
- **Accessibility fix found along the way:** the preview banner was a nested `<aside>` landmark (present before this work); it is now a note.

Screenshots (real local backend, seeded test accounts)

Student home:


py
Python Academy
Courses
Small steps. Real skills.
Sign out
YOUR LEARNING
Welcome back. Keep building.
UP NEXT
Meet the player
Python Foundations · Unit 1: Build Your Own Adventure
Continue →
200 XP
2 badges earned
All courses
AVAILABLE
1 of 12 units open
Python Foundations
Write your first programs, then grow them into tested, organized projects.
2 of 5 lessons complete
COMING SOON
Hackathon Ready
Scope, build, and demo a small working project with a team, against the clock.
Builds on Python Foundations units 1–7.
COMING SOON
Git and GitHub Essentials
Track changes, work on branches, and open your first pull request.
Builds on Python Foundations unit 1.
COMING SOON
Web Foundations: HTML and CSS
Build an accessible, responsive personal website.
Builds on Python Foundations units 1–3.
COMING SOON
JavaScript and the DOM
Make web pages interactive with events and state.
Builds on Python Foundations units 1–5 and Web Foundations: HTML and CSS.
COMING SOON
Data and SQL
Ask questions of real tables with queries, joins, and constraints.
Builds on Python Foundations units 1–4.
COMING SOON
Data Investigation with Python
Answer an original question with a documented dataset.
Builds on Python Foundations units 1–9 and Data and SQL.
COMING SOON
Computer Science Foundations
Representation, searching, sorting, complexity, and recursion.
Builds on Python Foundations units 1–5.
COMING SOON
Node.js and APIs
Build an API with routes, validation, and a database.
Builds on JavaScript and the DOM and Data and SQL.
COMING SOON
Full-Stack Builder
Ship a client, API, and database project with automated checks.
Builds on Git and GitHub Essentials and Node.js and APIs.
COMING SOON
Engineering Depth
Electives in C and memory, discrete foundations, numerical methods, and embedded simulation.
Builds on Python Foundations and Computer Science Foundations.
▶ Your account history
Built for curious minds. Designed for room to grow.

Course page:

py Python Academy Courses Small steps. Real skills. Sign out

Courses › Python Foundations

COURSE

Python Foundations

Write your first programs, then grow them into tested, organized projects.

2 of 5 available lessons complete

Continue: Meet the player →

UNIT 1 — 2 of 5 complete

Build Your Own Adventure

Output, variables, input, and choices in one connected story.

- ✓ **Open the story** — Output · print() · Story spark badge — Completed
- ✓ **Create your hero** — Variables · strings · Character creator badge — Completed
- 3 **Meet the player** — Interaction · input() · Conversation starter badge — Up next
- 4 **Choose a path** — Decisions · if / else · Path finder badge — Not started
- 5 **Launch your adventure** — Combine your skills · Adventure maker badge — Not started

UNIT 2 — COMING SOON

Numbers and decisions

Arithmetic, conversion, comparison, and Boolean logic.

UNIT 3 — COMING SOON

Repeat and explore

while and for loops that always finish.

UNIT 4 — COMING SOON

Keep track of a world

Lists, dictionaries, and sets.

UNIT 5 — COMING SOON

Organize your code

Functions, parameters, return values, and scope.

UNIT 6 — COMING SOON

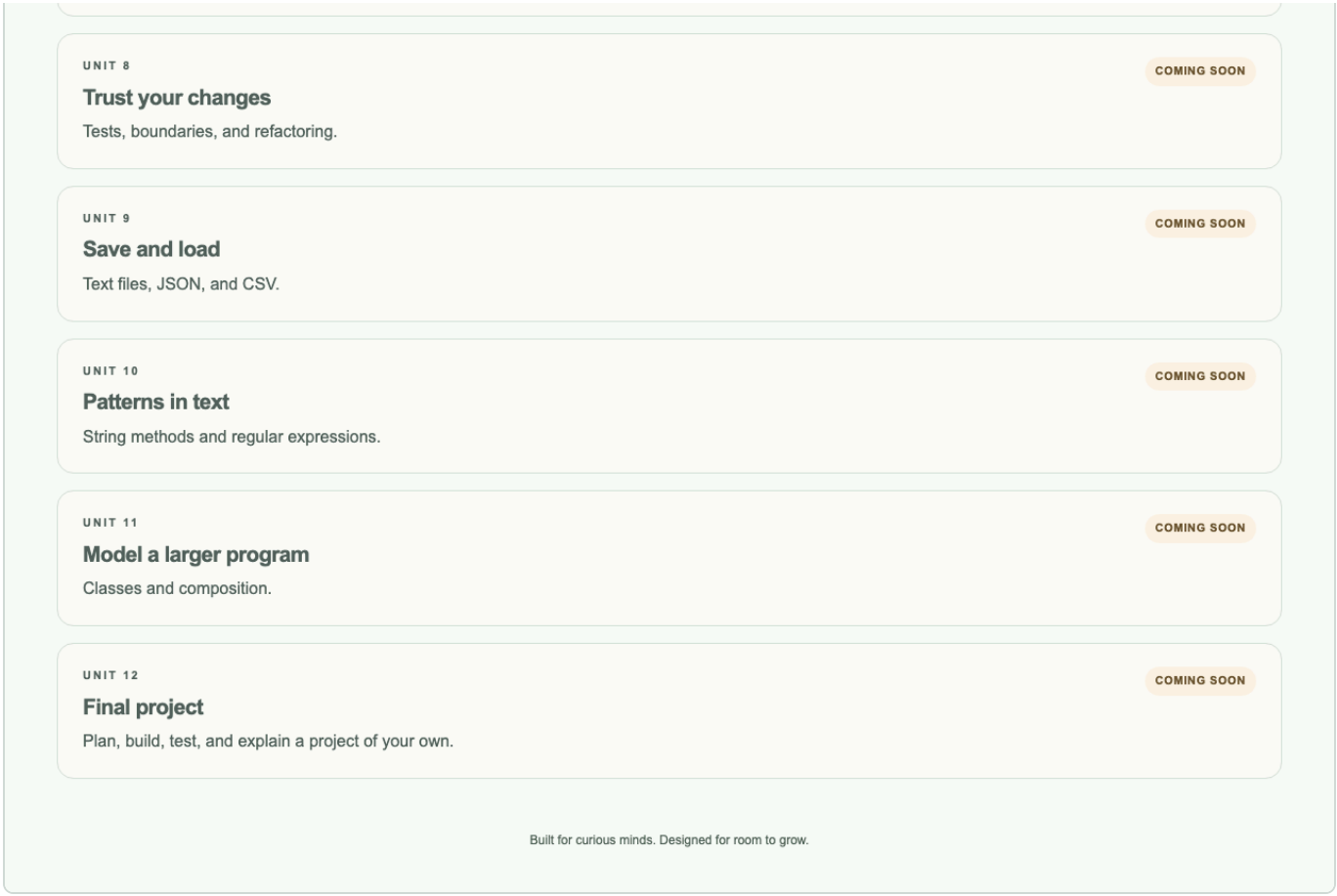
Recover and investigate

Tracebacks, debugging, and exceptions.

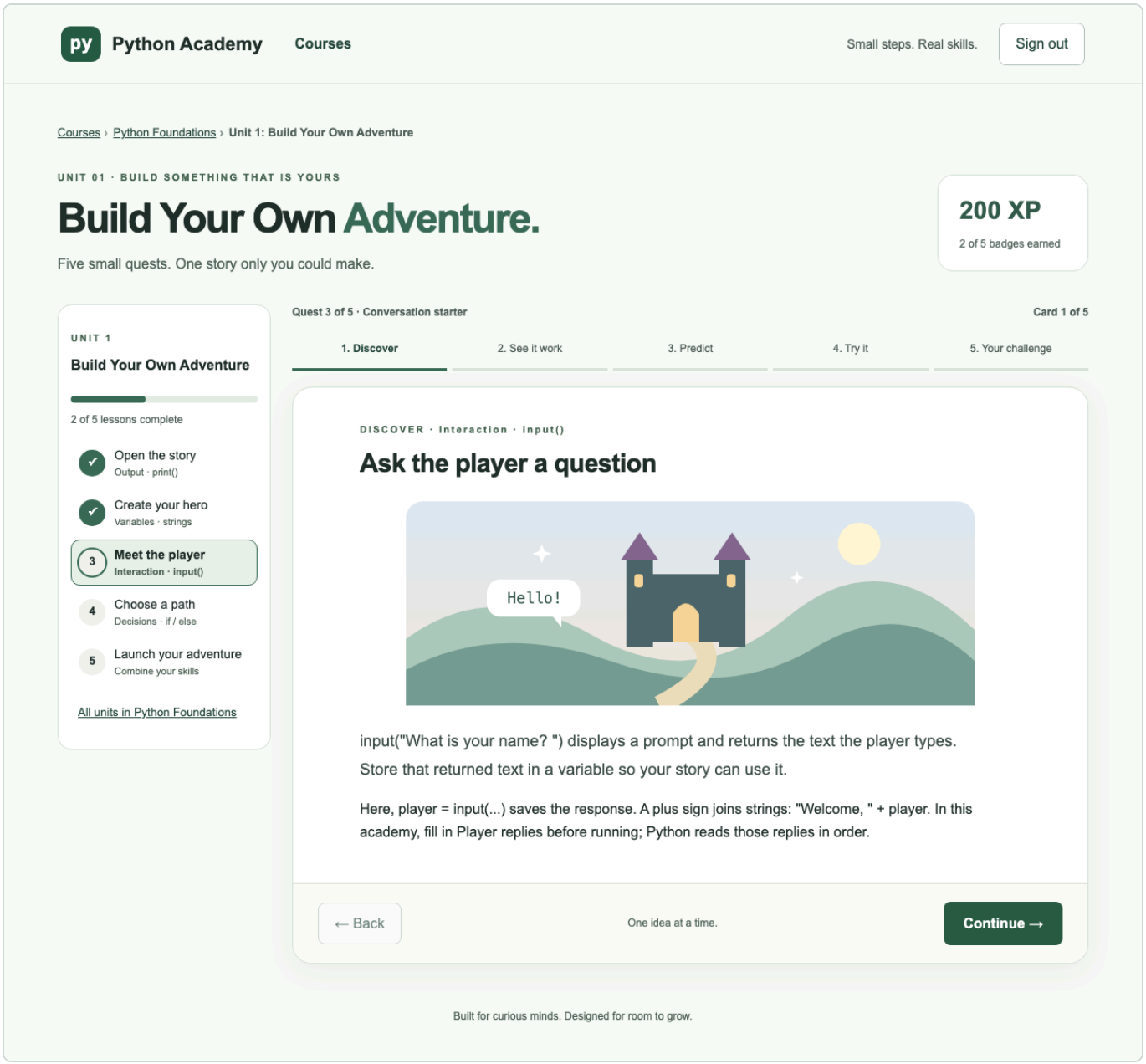
UNIT 7 — COMING SOON

Libraries and modules

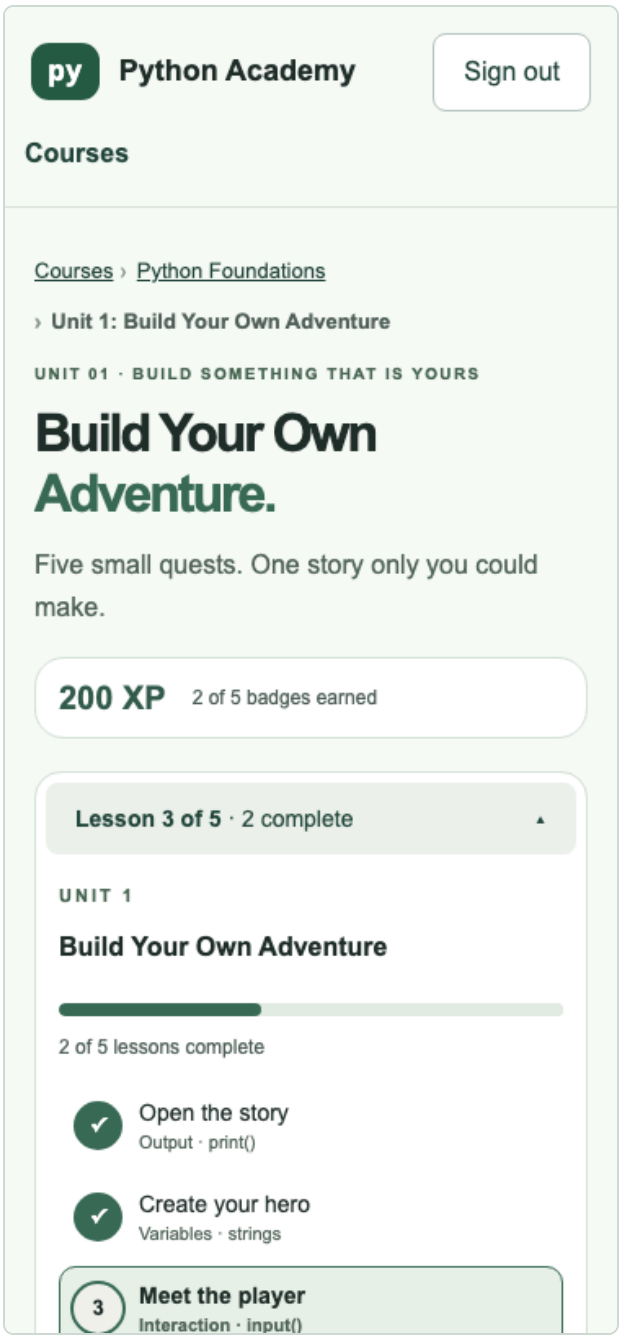
Importing and using the standard library.



Lesson page with unit sidebar:



Phone, lesson menu opened:



Coming-soon course (no start button):

COMING SOON

Hackathon Ready

Scope, build, and demo a small working project with a team, against the clock.

Builds on Python Foundations units 1–7
Helpful first Git and GitHub Essentials and Web Foundations: HTML and CSS

> This course is coming soon and cannot be started yet. Here is what it will cover.

UNIT 1 — COMING SOON

Idea to MVP

Cut an idea down to a minimum viable product.

UNIT 2 — COMING SOON

Time-boxing

Plan in fixed time blocks and know what to drop.

UNIT 3 — COMING SOON

Teamwork with Git branches

Feature branches, small pull requests, and reviews.

UNIT 4 — COMING SOON

Using public APIs safely

Rate limits, terms of use, and keeping keys out of frontend code.

UNIT 5 — COMING SOON

Quick UI prototyping

From a paper sketch to a minimal working interface.

UNIT 6 — COMING SOON

Demo and pitching

A two-minute demo that shows what works.

UNIT 7 — COMING SOON

Judging criteria

How projects are judged, and assessing your own honestly.

Acceptance criteria

- Catalog with course cards: title, short description, progress, "Coming soon" labels. **Met.**
- Course page with units and lessons. **Met.**
- Lesson page keeps one card at a time, with a sidebar showing done/current/not started; collapsible on mobile. **Met.**
- Existing visual style, accessibility, keyboard, reduced motion, responsive layout. **Met in tests and checks below.**
- Coming-soon courses not startable. **Met:** no start or lesson links (tested).
- Backfill re-run and read endpoints first. **Met:** separate first commit.

Tests run

Local: Python 3.12.14, Node 24.19.0, PostgreSQL 16.2 (throwaway `pgserver`).

Command	Before (`main`)	After
`cd backend && python -m pytest -q` (SQLite)	80 passed, 7 skipped	**84 passed, 8 skipped**
`cd backend && TEST_POSTGRES_URL=… python -m pytest -q`	87 passed	**92 passed**
`cd frontend && npx tsc --noEmit`	—	**no errors**
`cd frontend && npm run build`	succeeded	**succeeded** (new routes `/courses`, `/courses/[course]`, `/learn/[lesson]`)
`cd frontend && npm run test:e2e`	15 passed	**19 passed**

New or changed tests:

- **Backend:** catalog is public and matches the fixture; awards are student-only and correct over repeated passes; progress and the parent's completed list come from the ledger (legacy arithmetic excluded); `0004` awards a pass saved by pre-ledger code after `0003`.
- **Browser:** catalog shows every course with coming-soon labels and no way to start them; course page shows statuses and Continue goes to the right lesson; sidebar shows done/current/not started and works from the keyboard (focus + Enter moves to the lesson and focuses its heading); student home shows next lesson and XP; mobile menu starts collapsed, opens, stays inside 375px, and closes after choosing a lesson; reduced motion stops animations.

Accessibility check: axe-core 4.10.3 (WCAG 2.0/2.1/2.2 A and AA plus best practices) on 10 pages × desktop and phone, run against the real stack: **0 violations** after the landmark fix.

Real stack: uvicorn plus `next start` with seeded parent and student accounts. No page errors and no horizontal scroll on any page at 1280px or 375px.

Mocks and unverified behaviour

- Browser tests mock the API from the frozen fixtures; Python execution in them is real.
- Screenshots and axe used a real local backend with locally signed test sessions, not Google sign-in.
- **Not verified:** the deploy on Render, and real Google sign-in through the new pages. Not tested on Safari or Firefox (tests use Chromium).

Known limitations

- The lesson page still loads the legacy `/api/lessons` shape for unit 1; new units will need the lesson page to read lessons per unit.
- "Up next" is simply the first unfinished lesson; goal-based recommendations come in stage 4.
- No screen-reader testing with a real screen reader (only automated checks and accessible-name assertions).

After merging: checking the live site

1. Render → **python-academy-api** → **Logs**: `Database schema is at migration 0004`.
2. Open https://python-academy-dji8.onrender.com/courses: Python Foundations "Available", the rest "Coming soon".
3. Open `/courses/hack`: no start button.
4. Sign in as a student: home shows Up next; open a lesson and check the sidebar; on a phone, the lesson menu opens and closes.

Next concrete action

Maintainer reviews this PR. After merge and a live check, start stage 4: goal card at sign-up and the rule-based recommendation function with tests.